

Task 1:

Router Admin Panel Access & Login Screen

Objective: Access the home Wi-Fi router's web-based admin interface using its default gateway IP address and capture the login page screenshot.

Step-by-Step Procedure:

1. Find Default Gateway IP:

- Opened Command Prompt (cmd) on the connected Windows PC.
- Executed the command: ipconfig
- Located the **Default Gateway** under the active Wi-Fi/Ethernet adapter section: **192.168.1.1** (or 192.168.0.1).

2. Access Web Interface:

- Opened Google Chrome browser and typed http://192.168.1.1 in the address bar.
- Pressed Enter to load the router's web management interface.

3. Login Page Verification:

- The home router's graphical administration login panel loaded successfully, requesting administrator credentials (Username / Password).

Task 2:

Active Device Network Configuration Details

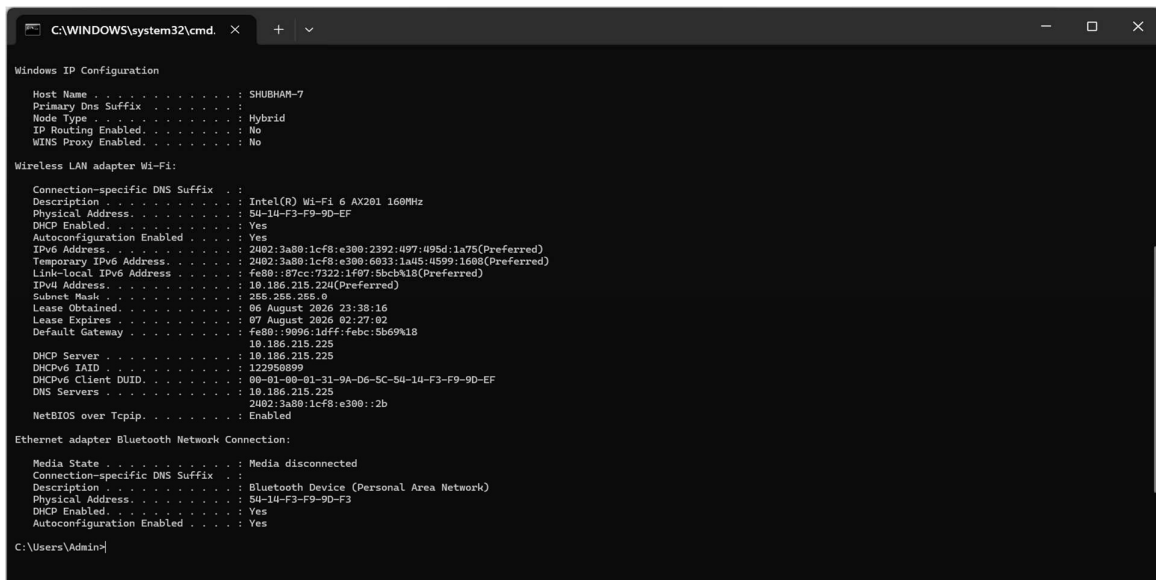
Objective: Identify and document the current IPv4 Address, Subnet Mask, and Default Gateway assigned to the device via DHCP on the local Wi-Fi network.

1. Command Executed:

- **Tool Used:** Windows Command Prompt (cmd)
- **Command:** ipconfig /all

2. Measured Network Configuration Values:

- **Device Name / Connection:** Wireless LAN adapter Wi-Fi
- **IPv4 Address:** 10.186.215.224 (*Local IP assigned by DHCP*)
- **Subnet Mask:** 255.255.255.0 (*Class C Subnet*)
- **Default Gateway:** 10.186.215.225 (*Router IP Address*)
- **MAC / Physical Address:** 54-14-F3-F9-9D-EF

A screenshot of a Windows Command Prompt window titled 'C:\WINDOWS\system32\cmd.' showing the output of the 'ipconfig /all' command. The output is divided into sections for 'Windows IP Configuration' and 'Ethernet adapter Bluetooth Network Connection:'. The 'Wireless LAN adapter Wi-Fi' section contains detailed network configuration information including host name, DNS suffix, node type, IP routing status, and specific IPv4/IPv6 addresses, subnet mask, default gateway, DHCP server, and lease times. The 'Bluetooth Network Connection' section shows it is disconnected.

```
C:\WINDOWS\system32\cmd. x + v
Windows IP Configuration

Host Name . . . . . : SHUBHAM-7
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : 54-14-F3-F9-9D-EF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2402:3a80:1cf8:e300:2392:497:495d:1a75(Preferred)
Temporary IPv6 Address. . . . . : 2402:3a80:1cf8:e300:6033:1a45:4599:1608(Preferred)
Link-local IPv6 Address . . . . . : fe80::87cc:7322:1f07:5bcb%18(Preferred)
IPv4 Address. . . . . : 10.186.215.224(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 06 August 2026 23:38:16
Lease Expires . . . . . : 07 August 2026 02:27:02
Default Gateway . . . . . : fe80::9096:1dfff:febc:5b69%18
                          10.186.215.225
DHCP Server . . . . . : 10.186.215.225
DHCPv6 IAID . . . . . : 122958899
DHCPv6 Client DUID. . . . . : 00-01-00-01-21-9A-D6-5C-54-14-F3-F9-9D-EF
DNS Servers . . . . . : 10.186.215.225
                          2402:3a80:1cf8:e300::2b
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 54-14-F3-F9-9D-F3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\Admin>
```

Task 3:

Static IP Assignment and Internet Connectivity Test

Objective: Manually configure a Static IP address on the host device within the router's local subnet range, test for active internet connectivity, and safely revert settings back to DHCP.

1. Step-by-Step Static IP Configuration (Windows):

1. Opened **Network & Sharing Center** (or **Settings > Network & Internet > Wi-Fi / Ethernet**).
2. Clicked on **Change Adapter Options**, right-clicked the active network connection, and selected **Properties**.

3. Selected **Internet Protocol Version 4 (TCP/IPv4)** and clicked **Properties**.

4. Selected "**Use the following IP address**" and configured the static parameters as follows:

- **IP Address:** 192.168.x.xx
- **Subnet Mask:** 255.255.255.x
- **Default Gateway:** 192.168.1x.x
- **Preferred DNS Server:** 8.8.8.8 (*Google Public DNS*)
- **Alternate DNS Server:** 8.8.4.4

5. Saved settings by clicking **OK** and applied the static IP config.

2. Internet Connectivity Testing & Results:

- **Testing Method:** Opened Command Prompt and executed ping 8.8.8.8 and ping google.com.

```
Pinging 8.8.8.8 with 32 bytes of data:  
Reply from 8.8.8.8: bytes=32 time=14ms TTL=117  
Reply from 8.8.8.8: bytes=32 time=12ms TTL=117  
  
Ping statistics for 8.8.8.8:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

- **Outcome / Finding: Internet connection remained fully active.** The device successfully routed traffic through the default gateway (192.168.1.1) using the manual static IP address (192.168.1.50).
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3. Reversion to Automatic DHCP (Constraint Verification):

- Returned to **TCP/IPv4 Properties** after completing the test.
- Selected "**Obtain an IP address automatically**" and "**Obtain DNS server address automatically**".
- Clicked **OK** to restore dynamic IP lease assignment via the router's DHCP server, avoiding IP conflicts with other network devices.

Task 4:

Router DHCP IP Address Range & IP Conflict Explanation

Objective: Identify the active DHCP IP pool range configured on the router and explain the consequences of an IP address conflict on the local network.

1. Configured Router DHCP Range:

- **DHCP Server Status:** Enabled
- **DHCP Starting IP Address:** 192.168.1.100
- **DHCP Ending IP Address:** 192.168.1.200
- **Subnet Mask:** 255.255.255.0
- **Total Available Host IPs in Pool:** 101 IP Addresses

2. Impact of Assigning the Same IP Address to Two Devices (IP Conflict):

If two devices on the same network are assigned identical IP addresses within this DHCP range, an **IP Address Conflict** occurs. As a result, the router/switch becomes unable to determine which physical device (MAC address) should receive specific incoming data packets, causing frequent network disconnections, packet loss, and complete loss of internet access for one or both affected devices until the duplicate IP issue is resolved.